

ABRA — Technical Documentation

Version 3.59.0 · Last updated 2026-08-06

Change record for 3.59.0. The headline metric changed. It was the win rate. It is now the exploitability. Read ADR-003 for the decision. Read `docs/POKER-TO-POKEMON.md` for the theory.

The reason is a measurement from other persons. VGC-Bench is the only published work in this format. Its authors trained a policy on more than 700,000 human battle logs. They then improved it with PPO, self-play, fictitious play and double oracle. The policy defeated a World Championships competitor in a mirror match. The authors then trained a best response against each of their own agents. Almost all of the agents were approximately 100% exploitable. Their expert tester wrote that strong human players adapt and defeat the agent after sufficient successive games.

A fixed policy is a map from a state to an action. In a game with hidden information, a best response can find the weak states of such a map. This project therefore tests one claim: an agent that computes a new answer each turn is more difficult to exploit than an agent that recalls a stored answer. **This claim is not proven. It is the experiment.**

The speed figure also changed. ADR-001 recorded 29 against 3,401 battles/sec/core. It gave a ratio of 117x. The measurement was made again on this machine. Both engines used the same four teams. The teams are derived from the store. Each run was 8 seconds with a 60-turn limit. MEDICHAM gives 13,041 turns/sec and 217 battles/sec. `champions_sim` gives 523 turns/sec and 28 battles/sec. The ratio is 24.9x. Use `turns/sec`. Do not use `battles/sec`: MEDICHAM ran to its 60-turn limit and Showdown ran with `choose('default')` to a natural end, so a battle is not the same quantity of work in the two engines. The old figures stay in ADR-001. The decision in ADR-001 stays correct. The stated reason for that decision changed: the engine is justified if, and only if, the search gives a measured gain.

The work is now in four phases. Complete MEDICHAM. Then run gate ROADMAP #62. If the gate passes, build the search and measure the exploitability against approximately 100%. If the gate does not pass, use the method of VGC-Bench: behaviour cloning, then PPO with self-play, fictitious play and double oracle. Phase 4 is a result. It is not a failure.

Change record for 3.47.0. Two artifacts were computed from an engine that then changed. Do not quote such an artifact. First measure whether the change moved the feature function: run every feature column through the old engine and the new engine on the same rows. All 58 columns were identical on 1,751,688 rows from 9,230 games. Then the two artifacts were computed again with the new engine. The result is the same: the model puts less probability on an action no human chose, **-0.002613, 95% CI [-0.003650, -0.001672]**, on 48,274 held-out decisions. `board.js` gives the priority rule a body with no type list. This is wrong for 0 of 1,751,688 rows. Do not change it yet.

Change record for 3.42.0. The fit checks whether a recorded action was a click. 1,336 of 241,927 actions were not clicks. Encore replaced 1,116 of them. A phazing move dragged in 220 of them. These actions are removed from the training labels. They are counted. 3,260 redirected attacks are kept. They are fitted over a set of two possible targets, not one certain target. The estimator was tested on data with known answers first. It recovers 97.4% of a planted error. The model now puts less probability on an action no human chose: **-0.002614, 95% CI [-0.003647, -0.001610]**. The redirection change did not improve anything that can be measured. The overall top-1 accuracy did not change. Two budget counters were replaced. The old counter measured two different things at once.

Change record for 3.45.0. The matrix did not test a move that can miss. It also did not test a move with a chance side effect. Many of these are the moves that players use most. The rule was wrong. Both engines use the same fixed die, so a miss happens in both engines and cancels. The removed counts are in `CHANGELOG.md`.

The two dice were not the same die. The harness set one die to the middle value. It set the other die to a different rule. Showdown checks accuracy with the second die. So every move below 100 accuracy missed in the official engine and hit in the simulator. This made false disagreements. It also made many cases look empty. Nobody saw this, because all of these moves were removed from the test first.

The harness now checks that the two dice agree. The check runs when the file loads. The matrix stages 2,300 pairs of 8,795. At this release 1,453 of them could occur, and the matrix agreed with the official engine on 1,436 of these. Release 3.46.0 changed both of those counts. Twelve new faults in the simulator are recorded. They are not repaired in this release.

Change record for 3.46.0. A test needs the other Pokémon to do something while it is hit. The harness gave it a move for this. If the Pokémon could not learn one of six safe moves, the harness gave it Protect. Protect stops the move under test. The test then showed no result. A test with no result looks the same as a test that cannot work. 379 of 2,300 tests were built this way.

The harness no longer uses Protect for this. It uses any move that the Pokémon aims at itself and that does not block. If no such move exists, the test is not built, and the reason is recorded. A check stops the run if a Pokémon is given a Protect for this purpose.

From `data/interaction-matrix.json` : 1,643 of the 2,300 tests can occur, and the matrix agrees with the official engine on 1,624 of these. The artifact also records 53 more disagreements in buckets the gate discards. These are real. Read `off_gate` and `off_gate_rows`, not the agreement rate alone.

Many tests that showed no result now show one. The count of these is in `CHANGELOG.md`. Three new faults in the simulator are recorded. One new fault in the harness is also recorded. The harness fault must not be counted against the simulator.

Change record for 3.44.0. Psychic Terrain stops a fast move only if the target stands on the ground. The simulator stopped the move against every target. A target that flies is not on the ground. A Flying type, a Pokémon with Levitate and a Pokémon with an Air Balloon all fly. A Pokémon that holds an Iron Ball is on the ground, even if it flies. The simulator now has one function for this question. Four parts of the simulator ask that function. Before this release three parts each had their own copy of the rule, and the copies did not agree. Grassy Terrain also used a copy. That copy healed a Pokémon with Levitate. Grassy Terrain must not heal a Pokémon that flies. All expected results come from the official engine. The count of tested mechanics is 210 of 213.

Change record for 3.43.0. The interaction matrix now checks its own arithmetic. The rule is `theoretical = staged + dropped`, for each axis. The generator stops if the rule is broken. The rule found three faults. The count of theoretical pairs was too small by 170. The depth limit on the type axis lost one pair each time it applied, 32 times. Four cases were counted in two result groups. After the repair the matrix stages 1,675 pairs of 8,676, and 1,031 of them can occur. The matrix agrees with the official engine on 1,027 of these 1,031 cases.

The agreement figure is lower than in 3.41.0. The simulator did not get worse. The earlier figure of 899 of 899 was measured on a smaller set. The four cases that disagree were not made before this release.

Change record for 3.41.0. The simulator covers 202 of 205 probed mechanics. Three are missing. Each has a written reason. The interaction matrix agrees with the official engine on 899 of 899 cases. The pair-layer fit now uses all four team-sheet channels (95,886 turns, 99.7% channel reach). The value of the sheet was measured on 44,982 held-out decisions: the likelihood gain is real; a top-1 gain is not shown. The pory family was refit on the current corpus. The DEAD-tag count fell from 61 to 38.

Change record for 3.40.0. The simulator covers 181 of 186 probed mechanics. Five are missing. Each has a written reason. The two rulebook files were compared. 151 facts were comparable. Two facts did not agree. The engine now reads the format's own secondary-effect chance. The MAG fit now uses all four team-sheet channels. A counter proves the channels reached the board (99.67% of 231,722 decisions). The pair-layer fit does not use them yet. The coverage plan changed: mutation tests come before the handler registry. See `COVERAGE-PLAN-REVIEW.md` for the reasons. ABRA has no exploitability number.

Change record for 3.56.0. The simulator did not use the accuracy of a move correctly. Four rules change accuracy. The rules are Coil, Wide Lens, Sand Veil and No Guard. We measured each rule two times. The first measurement had the rule. The second measurement did not have the rule. The two measurements were the same for all four rules. This shows that no rule had an effect.

There are three causes. The first cause is a table that changed the words `accuracy` and `evasion` into an empty value. Eleven parts of the simulator read that table. The second cause is that the simulator did not read items and did not read abilities for accuracy. The third cause is that the function which makes the accuracy decision does not receive the attacker and does not receive the target. A function cannot use an item on a body that it does not have.

The simulator now has one function for accuracy. The name of the function is `hitChance`. The function receives the attacker, the target, the move and the field. Four parts of the simulator call this function. The accuracy decision now happens after the target is known.

The simulator also did not make a Substitute. It removed one quarter of the health of the user. It did not give a Substitute to the user. A player made this move 1,976 times. The move was worse than doing nothing. The simulator now gives a Substitute. A second use of the move fails and does not remove health. One function decides if a move goes through a Substitute. The damage rules and the status rules both use that function.

The count of tested mechanics is 231 of 232. One mechanic is missing. The mechanic has a written reason. Five more rules are known to be absent. Each of the five has a written reason and a usage count. The largest is Aura, which needs information the damage function does not receive.

Change record for 3.50.0. The simulator did not do Taunt. It recorded the Taunt condition. It did not read the condition. A Pokemon with Taunt could still use a status move. The simulator now refuses a status move at two times. The first time is move selection. The second time is move execution. The rule comes from the tag data. It does not use the move name.

A pivot move had the priority of a switch. Parting Shot moved before all other moves. This is not correct. A pivot move is a move. It now uses its own priority.

Volt Switch changed the user when the target absorbed the move. This is not correct. The user now stays if the move does no damage.

Yawn worked against Good as Gold. This is not correct. Good as Gold refuses all status moves from an opponent. The Yawn code now asks.

The count of known differences with the official engine went from 94 to 72.

Change record for 3.49.1. The mutation harness reported 97 defect candidates. The two largest were not defects. The engine reads the Life Orb tag for the damage. The engine uses the item name only for the recoil. This is latent. The engine ignores the Light Screen tag value on purpose. The tag holds the singles value. This is a doubles engine. The doubles value is different. The harness now grades each open operator. It uses four classes. Class A means no lookup and no name branch. Class B means the engine substitutes its own value. Class C means the engine uses the name. Class D means the test could not move the value. The grade comes

from a parse of the engine source. It does not come from a comment. No defect candidate is class A. The ratchet counts class A only. Class A is 163 operators. Three known cases test the rule. The harness stops if the rule fails these cases.

Written in ASD-STE100 Simplified Technical English. Sentences are short. The voice is active. One word has one meaning. The document follows the Diátaxis structure: Tutorial, How-to, Reference, Explanation.

1. Tutorial — run ABRA for the first time

Do these steps in order.

1. Get the code. Clone the repository `github.com/willhoop/abra` .
2. Get the sibling engine. Clone `github.com/willhoop/chomp` next to it. ABRA reads it at `../CHOMP` .
3. Open the site. Open `web/index.html` in a web browser. The site needs no build step and no server.
4. Visit a model. Click a house in the town. Open **PORY** and move the sliders. The win % updates.
5. Run one check. In a terminal, run `node engine/validate_damage.js` . It confirms the damage engine.

You now have the site and one validated model.

2. How-to — common tasks

Repair the raw archive before any reparse. `MODE=backfill node engine/durable-ingest.js data/games.ladder.jsonl` The hourly Action appends to the store while the raw archive is gitignored, so CI-collected games have no local raw log. `MODE=reparse` REFUSES to run while any stored game is missing one, because reparse rebuilds the store from the archive and would delete them.

Read which configuration produced a result (added 3.33.0). `node engine/run_stamp.js --show data/rollout-r3.json` Every gate artifact has a `<name>.meta.json` file beside it. The file records the rollout budget, the exploration rate, the horizon, content digests of every source the gate reads, the commit, and whether the working tree was dirty. `node engine/status.js` prints the headline of that file under the gate. Read three fields before you quote a result. `reconstructed: true` means the stamp was inferred from a commit and not observed; read `confidence` beside it. `git.dirty: true` means the commit does not describe what ran; use `source_digests` instead. `source_digests` holds hashes of working-copy bytes and `git.blobs` holds git object names. Do not compare the two. On Windows they differ because git changes the line endings.

Do not quote an engine-speed figure from a document (added 3.59.0). There is no script in this repository that measures the speed of the two engines. Three figures are on record for MEDICHAM — 3,401 battles/sec in ADR-001, 1,606 battles/sec in ROADMAP #61 and 13,041 turns/sec in the 3.59.0 correction. No artifact holds any of them. No test compares them. No ratchet fails when one of them moves. If you need a speed figure, measure it, record the method beside it, and state the unit: `turns/sec` compares the two engines and `battles/sec` does not, because the two engines end a battle under different rules.

Write a stamp from a new measurement (added 3.33.0). Call `require('./run_stamp.js').writeStamp({...})` at the point the run writes its numbers. Do not write a second sidecar format. One artifact recorded a probability column and did not record the exploration rate that produced it. A file written at rate 0 and a file written at rate 1 were then identical byte for byte, and they differed by almost four accuracy points. The published result could not be recovered.

Show the project state on a web page (added 3.32.0). `node web/build-status.js` then open `web/status.html` . The build step writes `web/status-data.js` . The page reads a script-tag global. Do not change it to `fetch()` . A `fetch()` of a local file fails under `file://` and shows no error to the reader.

Show the model select screen (added 3.32.0). Open `web/stadium.html`. Run `node tests/test-stadium-roster.js` after you add or remove a model. The test compares the page against `docs/MODELS.md`. The test fails if a model has no cabinet.

Compare the engine with Showdown (changed 3.32.0). `SHOWDOWN_PATH=... node tests/test-engine-diff.js --seed 20260804` The sampler is seeded. Two runs with the same seed give the same result. Before 3.32.0 the sampler used `Math.random()` and the count changed between runs. Always record the seed with the count.

Check that every room parses (added 3.34.0). `node tests/test-web-parses.js` This test runs the inline script of each page through a parser. A page can contain correct text and still fail to run. On 2026-08-04 a page had one wrong quotation mark. The page showed only its title. Two other tests gave the page full marks, because they read the page as text.

Check that the live site has every room (changed 3.34.0). `node tests/test-site-sync.js` `app/` is the folder the web server uses. `web/` is the folder you edit. The test now compares EVERY page in `web/` against `app/`, and also the data files that a page loads. Before 3.34.0 it compared one file only, so a NEW page that was never copied stayed invisible. If the test fails, run `cp web/<page> app/<page>`.

Check whether an artifact is too old to trust (changed 3.34.0). `node engine/provenance.js` The report gives a drift percentage and, beside it, `ci_gain` and `max_shift`. Use `max_shift`. It states how far the missing games can move the result. A percentage of a store that grows every hour only states the AGE of the artifact. No artifact in this project can move a proportion by one percentage point.

Run the official Champions engine. `SHOWDOWN_PATH=/path/to/pokemon-showdown node engine/champions_sim.js` Needs a BUILT master checkout; the champions mod is not in the npm package.

Generate self-play games (MEW). `SHOWDOWN_PATH=... node engine/mew.js --n 1000` then `SHOWDOWN_PATH=... node engine/validate_selfplay.js` Output goes to `data/games.selfplay.jsonl` and must NEVER be pooled with the ladder store.

Fit the scoring bot's policy. `SHOWDOWN_PATH=... node engine/fit_policy.js` Reads every clean open-team-sheet game, builds one row per human decision, fits by conditional logit, and writes `data/policy-weights.json`. It prints the held-out comparison against the behaviour clone; if the fit does not beat the clone it says so in plain words and the weights must not ship. `engine/selftest.js` refuses a weight file that lost to the clone, and refuses one whose feature list does not match what `engine/board.js` computes.

Check the corpus the policy was fitted on. `SHOWDOWN_PATH=... node engine/corpus_shift.js` Compares open-sheet against closed-sheet ladder on the same code: team composition, behaviour given a board, bot contamination before and after filtering, and mean rating. Run it before trusting a refit. The teams differ by 551.9 points of total absolute species difference; the measured behaviour differs by at most 1.49.

`fit_policy.js` then re-estimates on a sample reweighted to the closed-sheet species mix and reports the shift **in standard errors**. Five weights move materially — `priorLogP` by 10.8 SE, `bp` by 6.2 SE (its sign flips) — so the **reweighted vector ships**. Board-reading weights (`eff`, `immune`, `deadStatus`) do not move: reading the board transfers between the two metagames, how much popularity is worth does not.

Pull the Bo3 open-team-sheet ladder. `PAGES=6 CONC=20 FORMATS=gen9championsvgc2026regmbbo3 node engine/durable-ingest.js data/games.bo3.jsonl` CI does this hourly. This format's ruleset carries `Force Open Team Sheets`, so every game publishes all six sets of both sides — the only continuously-collected corpus in which the choice set of a decision is known. The main ladder format carries plain `Open Team Sheets`, which is optional and needs both players to agree, which is why only ~1% of the closed store has sheets. **It goes in its own store and is never pooled with the ladder store** — different information regime, different metagame. Dedupe it with `python3 engine/dedupe_store.py data/games.bo3.jsonl --write`.

Play with the scoring policy. `SHOWDOWN_PATH=... node engine/mew.js --n 1000 --policy score`

--policy	what it does
random	Showdown's <code>RandomPlayerAI</code> . Correct for plumbing and matchup structure; not valid as training data
prior	samples the move a species actually clicks, from <code>data/move-priors.json</code> . Board-blind
score	tracks the board and scores every (move, target) pair with the fitted weights. The only mode that aims

Check the run's own accounting: `policy=score` must report ~100% of decisions scored and a non-zero `aiming:` line. A run reporting 0% is not a scoring bot.

Compare two policies fairly. Pass the SAME `--seed` to both runs — MEW derives its team sampling from it, so the two corpora play identical teams and the comparison is paired rather than confounded by which teams each happened to draw:

```
node engine/mew.js --n 600 --policy prior --seed 4242 --out data/_a.jsonl
node engine/mew.js --n 600 --policy score --seed 4242 --out data/_b.jsonl
node engine/realism_report.js --self data/_a.jsonl
node engine/realism_report.js --self data/_b.jsonl
```

Generate self-play at scale (the farm). `SHOWDOWN_PATH=... node engine/mew_farm.js --n 200000 --procs 12 --conc 1`

`--conc` must stay at 1. The simulator is synchronous and CPU-bound, so in-process concurrency never overlaps real work; it only holds N battles live at once and multiplies GC pressure. Measured on 8 physical / 16 logical cores: 8 procs at `--conc 4` gave 11 games/sec, the same 8 procs at `--conc 1` gave 38. Process count is the unit; 12 procs / conc 1 reproduced at 44–46 games/sec.

Two files are written, and **both are needed**:

file	contains	read by
<code>data/games.selfplay.jsonl</code>	game-level records (six / brought / lead / winner)	store-shaped engines
<code>data/games.selfplay.raw-logs.jsonl</code>	full protocol logs, {id, uploadtime, log}	PORY, <code>state_encoder.py</code>

The value models reconstruct per-turn board states by replaying the **protocol log**; the records are summaries and contain no per-turn state. A corpus written without the sidecar is unreadable by the models it exists to train. Budget ~5 KB per game per file.

Every battle is replayable. A record carries `id`, `selfplay.seed`, `selfplay.policy` and `selfplay.engine_commit`, and both the battle dice and both players' decision PRNGs are derived from that seed. Re-running the same seed against the same pinned engine reproduces the game byte-for-byte (verified 25/25, excluding the `|t:|` wall-clock line). This is what makes a claim like "this switch won the game" checkable rather than asserted.

Teams are validated against Showdown, not against our own rules. `packTeam` enforces Item Clause during packing, then runs the official `TeamValidator` and repairs what it can; MEW discards anything still invalid rather than recording it. `BattleStream` does **not** validate — it plays whatever it is handed — so without this gate an illegal team produces a record indistinguishable from a legitimate game. Before the gate, the validator rejected 80.5% of the pool. Cost is 4.30 ms/team, ~9.6% of a battle.

Team preview is sampled, not fixed. `chooseTeamPreview` draws four of six weighted by measured $P(\text{brought} \mid \text{on team})$ and two leads weighted by $P(\text{lead} \mid \text{brought})$, from `data/bring-priors.json` (regenerate with `node engine/bring_priors.js`). `MEW_PREVIEW_TEMP` controls how far the draw wanders from the common line: 1.0 samples proportional to measured propensity, lower collapses toward the single most likely bring, higher flattens toward uniform.

Refresh the official priors. `node engine/fetch_smogon_stats.js` then `node engine/smogon_priors.js` CI does this on the 4th and 11th of each month.

Pull new replays. `PAGES=6 CONC=20 node engine/durable-ingest.js data/games.ladder.jsonl` The command adds only new games. It never duplicates a game and never re-fetches a stored game.

Rebuild the meta model. `node engine/analyze.js data/games.ladder.jsonl` writes `data/meta-usage.json`.

Refresh the site data. `python3 engine/refresh-site-data.py` writes `data/live.js`, `data/archetypes.json`, and `data/kad-replays.js`.

Run a model or an evaluation.

- GURU matchup matrix: `python3 engine/guru.py`
- XATU belief / policy eval: `python3 engine/eval_policy.py`
- PORY value net: `python3 engine/pory.py`
- CHOMP-EV proof: `node engine/chomp_ev.js`
- SLOWKING preview-Nash (species): `python3 engine/slowking_preview.py`
- SLOWKING preview-Nash (playstyle): first `node engine/playstyle.js`, then `MATRIX_FILE=data/playstyle-matchups.json TAG=playstyle python3 engine/slowking_preview.py`

Validate the damage engine. `node engine/validate_damage.js`. It fails if any scenario is more than 5% from the Smogon damage calculator.

Run the tests. `node tests/test-parse.js`, `node tests/test-dynamics.js`, `node tests/test-medicham.js`, `node tests/test-chomp-ev.js`, `python3 tests/test-jolteon.py`, `python3 tests/test-slowking.py`.

Edit the site, then mirror it. After you change `web/index.html`, copy it: `cp web/index.html app/index.html`.

3. Reference

3.1 Stored game record (`data/games.ladder.jsonl` , one JSON object per line)

Field	Meaning
<code>id</code> , <code>date</code>	replay id and upload time
<code>p1</code> , <code>p2</code>	<code>{name, rating, bot}</code> per player
<code>six.p1/p2</code>	the six revealed at preview
<code>brought.p1/p2</code>	the four actually brought
<code>lead.p1/p2</code>	the two led
<code>sets</code>	per species, the revealed moves / item / ability
<code>turns</code>	per-turn events (move, damage, faint, status, field)
<code>winner</code>	the winning name

3.2 Model outputs

File	Written by	Contents
<code>data/damage-validation.json</code>	<code>validate_damage.js</code>	damage error against the Smogon damage calculator
<code>data/guru-matchups.json</code> , <code>guru.js</code>	<code>guru.py</code>	archetype matchup matrix, Wilson CIs
<code>data/xatu.json</code> , <code>xatu.js</code>	<code>xatu.py</code>	opponent set/move belief
<code>data/pory.js</code> , <code>pory-eval.json</code>	<code>pory.py</code>	mid-game value net + its score
<code>data/chomp-ev.json</code>	<code>chomp_ev.js</code>	the bring proof (null result)

File	Written by	Contents
data/playstyle-matchups.json	playstyle.js	playstyle matchup matrix
data/slowking-eval.json , slowking-playstyle-eval.json , slowking*.js	slowking_preview.py	preview equilibrium + exploitability
data/policy-weights.json	fit_policy.js	the scoring bot's fitted weights, the feature list they were fitted against, and the held-out comparison against the behaviour clone
data/meta-usage.json , live.js	analyze.js , refresh-site-data.py	usage model + live site counts
data/protocol-events.json	derive_protocol_events.js	every event Showdown can emit, which of them <code>medicham2</code> emits, and a written reason for each one it does not

3.2a Protocol trace (`engine/medicham2-browser.js`)

The simulator can emit a Showdown-shaped protocol stream. It is **off by default**; nothing in the battle loop pays for it unless a caller asks.

To turn it on:

```
const trace = [];
const S = M.battleInit(teamA, teamB, { trace });
M.battleTurn(S, rng);
// trace is an array of protocol lines:
//   |move|p1a: incineroar|fakeout|p2a: garchomp
//   |-damage|p2a: garchomp|154/175
//   |cant|p2a: garchomp|flinch
```

To read it:

Export	Does
M.TRACE_EVENTS	the 36 event names this engine claims it can emit
M.traceCounts(lines)	counts by event name, PARSED from the lines rather than kept beside them
M.traceCanon(line)	the one normaliser — lowercase and strip whitespace per field

Identifiers are ids, not display names. The engine writes `p1a: incineroar` and `fakeout` where Showdown writes `p1a: Incineroar` and `Fake Out`. This engine holds no display-name table and inventing one would be a translation layer that can itself be wrong. `traceCanon()` is applied to **both** streams by any comparison driver, which makes the two agree by canonicalisation rather than by translation.

Do not add an event without regenerating the artifact. `node engine/derive_protocol_events.js --write` fails if a name in `TRACE_EVENTS` is one Showdown never emits, and fails if a Showdown event is neither emitted nor given a reason. `tests/test-protocol-trace.js` fails if a claimed event never fires in a real game.

3.3 Continuous collection

A GitHub Action (`.github/workflows/ingest.yml`) runs the pull hourly and commits the store. A separate tests workflow runs the test suite and the damage validation on every push and pull request.

4. Explanation

4.-2 Why the exploitability is the headline metric (added 3.59.0)

This section explains a decision. It does not give instructions. For the decision itself, read ADR-003.

There are two kinds of game. In the first kind, both players see all of the state. Chess and Go are of this kind. A search over that state is correct, and one best move exists at each position. In the second kind, each player holds private information. Poker is of this kind. VGC is also of this kind: you do not see which four of the six the opponent brings, the items, the abilities, or the fourth move of a set.

In the second kind of game, one best move does not exist. The correct object is a mixed strategy. If you always make the same choice in the same situation, an opponent who watches you can find that choice and defeat it. This is why the poker research of 2007 to 2021 produced CFR, DeepStack, Libratus and ReBeL, and not a larger chess search.

A fixed policy is therefore exploitable by construction. A behaviour clone is a fixed policy. A PPO agent is a fixed policy. Both give one answer for one state. VGC-Bench measured this on its own agents: it trained a best response against each agent and found approximately 100% exploitability, although one of those agents defeats a professional player. The two facts are consistent. Strength on average and readability under study are different quantities.

A win rate cannot show this defect. A win rate is measured against a population that does not adapt. An exploitability is measured against an opponent that is trained against you. Only the second measurement finds a policy that is strong today and readable after five games.

The claim under test in this project. A search that computes a new answer each turn shows no fixed map to an opponent. It should therefore be more difficult to exploit than a compiled policy. Three properties of VGC can defeat this claim, and all three are open:

property	why it can defeat the claim
simultaneous moves	sequential CFR does not apply directly to a turn node
stochastic resolution	damage rolls, accuracy and speed ties add variance to every leaf
short horizon	a median game is 6 turns, so there is little depth for a search to use

Do not read this section as a result. The project has no exploitability figure. `data/exploitability.json` is declared void. The comparison with approximately 100% is prepared and not yet made.

4.-1 Why additivity is the recurring failure (added 3.28.0)

The same defect has now appeared at three levels of this project, and it is worth stating once because the fix has the same shape every time.

A sum of independent terms cannot express a conjunction. If a score is $w_1*a + w_2*b$, there is no setting of w_1 and w_2 that makes "a AND b together" worth more than the parts. Not *scored badly* — **not representable**.

Where it bites:

level	the thing that cannot be said	consequence
one Pokémon (PORY, §4.0)	"material lead matters on turn 3, not turn 25"	the neural net exists for this
two Pokémon (MAG → DODUO)	"Protect with A while B removes the thing killing A"	independent per-slot choice; MAG aims both attacks at one foe ~50% of the time where humans do 23.4%
six Pokémon (JOLTEON → DITTO)	"Pelipper + Archaludon is worth more than the sum"	measured: the additive species block moves the score by ~6.3 while the two non-additive terms move it by ~0.4, so hill-climbing converges on the top-6 by weight

The literature agrees and names the boundary. Cooperative multi-agent value factorisation (QMIX, Rashid et al. 2018) constrains the joint value to be *monotonic* in each agent's utility, and QPLEX and Weighted QMIX exist precisely because that constraint cannot represent non-monotonic coordination. Our

case is the non-monotonic one.

But the expensive machinery does not apply here. That literature is about avoiding enumeration when agents are many. With **two** Pokémon the coordination graph is a single edge, so Variable Elimination degenerates to enumerating the joint actions and Max-Plus message passing is unnecessary. Measured on a real mid-game board: $9 \times 8 = 72$ joint actions per side, of which only **28** have a non-zero interaction term — the other 44 score exactly as the sum of two singles already computed.

And the fix is not "add the interaction term", it is "fit it for the right thing." DODUO's pair block exists and is fitted; it loses at 42.0% because it was fitted to *resemble human pairs* rather than to *win*. Expressiveness was necessary and not sufficient.

Practical rule. Before adding a feature, ask whether the thing you want to say is a conjunction. If it is, no weight on an individual term will ever say it — and if the model already has the interaction term, check what objective it was fitted to before concluding the idea failed.

4.0 What a neural network is here, and why it is not automatically an upgrade

Every model in ABRA answers one question: given the board right now, what is $P(I \text{ win})$? PORY answers it with **logistic regression** — multiply each feature by a weight, add them up, squash to 0..1:

```
p = sigmoid(w0 + w1alive_diff + w2hp_diff + ...)
```

That can only draw a straight dividing line through feature space. It cannot express "*being one Pokémon ahead matters enormously on turn 3 and barely on turn 25*", because that is an **interaction** between two features, and a sum of independent terms has no way to say it.

A **neural network** is the same idea with a middle step. Inputs are combined into H hidden units, each its own weighted sum passed through a nonlinearity (ReLU: $\max(0, x)$), and those are combined into the answer:

```
h = relu(W1 @ x + b1) # H learned intermediate quantities
p = sigmoid(W2 @ h + b2)
```

Each hidden unit can become a detector for a **conjunction** — "ahead on material AND my active is faster AND Trick Room is not up" — and the output layer weighs the detectors. Universal approximation (Cybenko 1989; Hornik 1991) says one hidden layer of sufficient width can represent any continuous function on a bounded domain, so the network is strictly **more expressive** than the linear model.

Strictly more expressive is a claim about representation, not about learning. Extra capacity spent on features that contain no interactions buys nothing and costs variance. `engine/pory_baseline.py` already established the relevant fact: PORY's six material features are **beaten by two of them** (`alive_diff + hp_diff` at 0.5822 vs PORY's 0.5840). If the features carry no more signal, a network fitted to them lands in the same place — and reporting otherwise would be measuring the estimator rather than the game.

This is the lesson the game-playing literature learned repeatedly. **TD-Gammon** (Tesauro 1994) needed hand-designed board features, not checker counts. **AlphaGo Zero / AlphaZero** (Silver et al. 2017, 2018) feed the value head a *stack of planes* — piece positions, repetition, side to move — because material is precisely the baseline a value net must beat, and it beats it by seeing **where** the pieces are. Leela Chess Zero's ablations show the value head collapsing toward the handcrafted evaluation when the positional planes are stripped.

So the binding constraint was **representation, not capacity**. `engine/state_encoder.py` supplies the planes: HP per slot, active vs benched, status, stat boosts, weather / terrain / Trick Room / Tailwind / screens, hazards, and the active Pokémon's types — 121 features where PORY had 6.

`engine/pory_nn.py` then separates the two explanations rather than conflating them, by running eight arms on one split:

arm	what it isolates
B2 logistic, <code>alive_diff + hp_diff</code>	the bar — two material features
L6 logistic, PORY's six	PORY itself
LR logistic, rich features	gain from representation alone
N6 network, PORY's six	gain from nonlinearity alone
NR network, rich features	both

If the network wins only on `N6`, the gain is nonlinearity. If only on `LR`, PORY was feature-starved rather than model-starved. If neither beats `B2`, that is the result and it is reported as such.

Species identity is deliberately **excluded**: ~240 species one-hot across 8 slots is a 1,920-dimensional input, which on ~9,000 real games would be fitted almost entirely to noise. Species enters only through type. Embeddings become defensible once the self-play corpus is large enough, and the encoder is versioned so that change is visible rather than silent.

Two methodological points that decide whether any of these numbers mean anything. **Splits are by game, never by state** — turns within a game share an outcome, so splitting on states leaks the label across the boundary and flatters every arm equally. And **every state is emitted twice with the sides swapped**, because antisymmetry in the two players is a property of the game; a model trained on p1's view alone will not respect it.

Store raw, analyse on top. ABRA saves every game with every fact it may ever need. Every filter — rating tier, humans-only, archetype, playstyle — runs over the store at read time. A change to how the games are segmented is a re-computation, not a re-download. This makes the fetch a one-time cost and keeps the analysis free to change.

Support decisions, do not predict outcomes. In this format, the winner of a game is near-impossible to predict from the two team sheets; even a player-rating model ties a coin. ABRA therefore judges each model on a decision, not on the match result. Each probability ships a proper score (log-loss or Brier), a confidence interval, and an honest baseline.

Why the confidence interval is clustered. States within one game are correlated. A confidence interval that resamples states would be too narrow. ABRA resamples whole games instead. For a matchup matrix, it also resamples each cell from its Beta distribution before it solves, so the interval carries the small-sample uncertainty.

Honest negatives are kept. Two results are negative and are reported plainly: the team-picker's brings do not beat a coin (CHOMP-EV), and the playstyle rock-paper-scissors cycle rests on small samples and is suggestive, not settled. A negative that is measured is more useful than a positive that is asserted.

Companion documents. [White paper](#) · [Deck](#) · [Project summary](#) · [Model ledger](#) · [Changelog](#)

How to measure against a frozen engine

Description

An engine release is a copy of every file whose content can change a measured number. A measurement reads the copy. Other work on the repository does not change the copy.

Twelve files are in a release: `medicham2-browser.js`, `board.js`, `rollout_leaf.js`, `position_features.js`, `tags.js`, `champions_sim.js`, `engine-data.js`, `abra-tags.js`, `tags.json`, `policy-weights.json`, `policy-weights-joint.json` and `move-priors.json`.

The release identifier is a digest of the file digests. If the files do not change, the identifier does not change. A second cut of the same files makes no second copy.

Procedure — cut a release

1. Make sure the files you want to freeze are correct.
2. Run this command. Give a reason.

```
node engine/engine_release.js cut "why this release exists"
```

1. Read the twelve digests in the output. Confirm the Showdown commit is not `UNKNOWN`.

Procedure — measure against a release

1. Open the release at the start of the measurement.
2. Load every engine file from the release, not from `engine/`.
3. Put the stamp in the artifact you write.

```
const REL = require('./engine_release.js').open();
const MEDI = REL.require('engine/medicham2-browser.js');
const artifact = Object.assign({}, REL.stamp(), result);
```

`REL.stamp()` writes `engine_release`, `showdown_commit` and `source_digests` into the artifact.
`engine/provenance.js` reads `source_digests` and compares the content. An artifact without this stamp can only be checked by timestamp, and a timestamp cannot show that a file changed after it was read.

Procedure — check a release

```
node engine/engine_release.js list          # each release, and how many files have moved since
node engine/engine_release.js verify <id>   # has the copy itself changed?
```

`list` shows the drift. Drift is normal. Drift tells you if an old number still describes the engine that ships today.

Warning

Do not read `engine/` files during a measurement. Another division can write them at any time. This does not cause an error. It causes a number that is wrong and looks correct.

On 2026-08-04 three divisions ran at the same time with separate files. The weights of the model under test changed between the two halves of one measurement. The measurement completed. No check failed. 7,100 games were lost.